

Gene Expression Matrix ( $M_{p,i}$ )

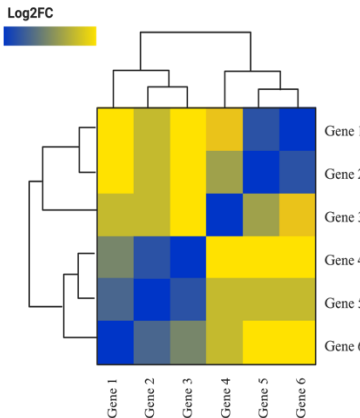


- Cancer Type
- Cancer Stage
- Treatment Response

$$\begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,i} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,i} \\ \vdots & \vdots & \ddots & \vdots \\ x_{p,1} & x_{p,2} & \cdots & x_{p,i} \end{pmatrix}$$

- Rows are patient ( $p$ ) and columns are genes ( $i$ ).

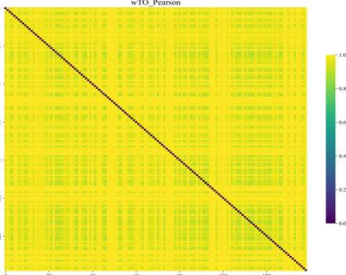
Preprocessing & Pre-selection ( $M'$ )



- Filtering
- Outlier Detection
- DESeq2

Co-expression Measures ( $C_{i,j}$ )

wTO (Pearsons)

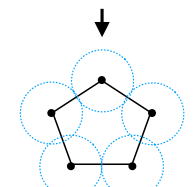


- $C_{i,j}$  is global co-expression measure for genes  $i$  and  $j$ .
- Use best-performing co-expression matrix from Clinical Outcome Prediction

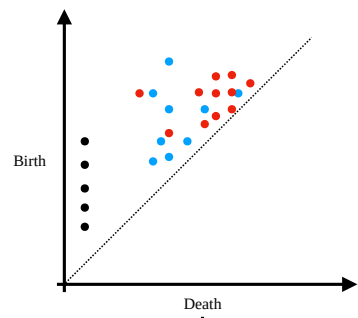
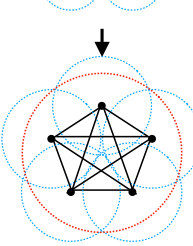
Persistent Homology

$$\begin{pmatrix} c_{1,1} & c_{1,2} & \cdots & c_{1,j} \\ c_{2,1} & c_{2,2} & \cdots & c_{2,j} \\ \vdots & \vdots & \ddots & \vdots \\ c_{i,1} & c_{i,2} & \cdots & c_{i,j} \end{pmatrix}$$

VR complex



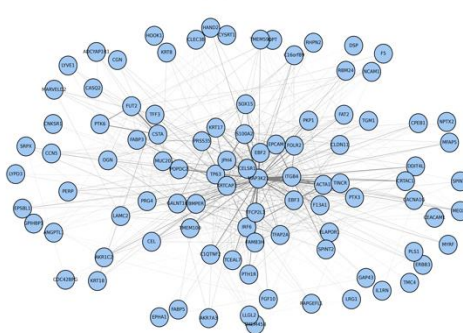
Persistent Homology



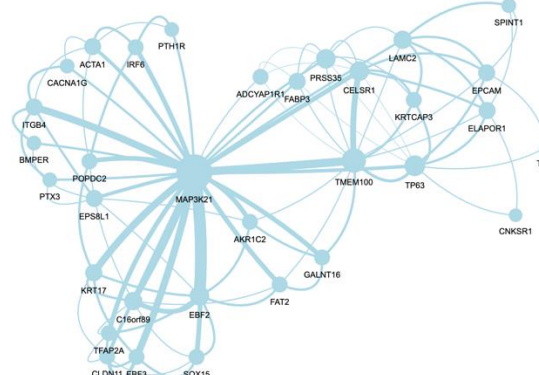
| $\beta$ | Gene set     | Lifespan |
|---------|--------------|----------|
| 2       | CDC42, PTH1R | 7.468    |
| 1       | MAP3K, ITGB4 | 0.234    |
| 2       | CDC42, BRCA  | 4.792    |

Network Visualization tool

Static



Interactive Web-based



Biomarker Discovery



- Gene signature that is validated through external literature